scanprobe

A read-only first-pass GPU evidence scan

`scanprobe` saves a few minutes in the standard GPU troubleshooting workflow.

Run it when a GPU node acts weird and you need the obvious local NVIDIA evidence before rerunning, draining, or filing a support ticket.

```
python3 scanprobe.py
python3 scanprobe.py --json
```

What It Checks

`scanprobe` gathers the first-pass checks an operator often runs by hand:

- `nvidia-smi` GPU visibility and basic GPU state
- ECC counters exposed by `nvidia-smi`
- temperature and hardware throttle flags
- current-boot NVIDIA Xid/kernel-log evidence when visible

It prints a primary issue, visible evidence, next action, and one conservative triage label:

- `CLEAR`: no local drain/watch evidence was visible
- `WATCH`: inspect visible evidence before rerunning expensive work
- `DRAIN`: do not launch new work until listed evidence is resolved
- `UNKNOWN`: this shell cannot see enough local GPU or kernel-log state

What It Does Not Check

`scanprobe` does not check silent data corruption, NCCL/fabric health, application correctness, cloud-provider state, storage, scheduler behavior, or performance regressions without visible local evidence.

It does not run DCGM, NCCL tests, CUDA samples, PyTorch, matmul tests, bandwidth tests, thermal stress tests, daemons, network calls, or telemetry.

It does not reset GPUs, change clocks, change persistence mode, drain nodes, or start a workload.

Why It Exists

In GPU incidents, the first pass often scatters across shell commands, screenshots, Slack threads, and support-ticket fragments. `scanprobe` puts that first pass into one read-only command and one pasteable report.

It answers one narrow question:

> Do local, visible NVIDIA GPU signals suggest this node is risky to use right now?

Alpha Ask

We need skeptical feedback from GPU infrastructure engineers on the ordinary workflow:

- real outputs from normal and weird nodes
- wrong verdicts
- confusing wording
- whether it saved time versus the manual first pass
- missing visible local evidence that would have changed the next action
- notes on where the command ran from: host, container, Slurm job, notebook, or Kubernetes pod

Success does not mean `scanprobe` diagnoses everything. Success means it saves a few minutes, catches obvious local evidence, and makes the first response easier to paste into Slack, Jira, or a provider ticket.

Repo: https://github.com/cv700/scanprobe